Kildisiute 2021 NB — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

2 populated samples (≥101 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached). Manifold resolves into two lobes; trajectory is scoped to the Embryonic lobe (1,504) — the Cycling lobe (216) confounds Gast with proliferation.

Gate 2 — module geography → GO (non-branching)

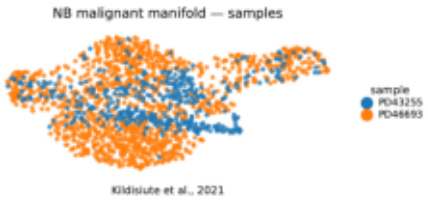
Gast/pSom co-vary (r=+0.91); along pseudotime Gast is flat (ρ=−0.01) and pSom declines (ρ=−0.26). Crest is anti-correlated with both (r=−0.66/−0.84), rising to a terminal readout (ρ=+0.40; neural-crest-high decile mean pt=0.70).

Gate 3 — hub mutual exclusivity → NA

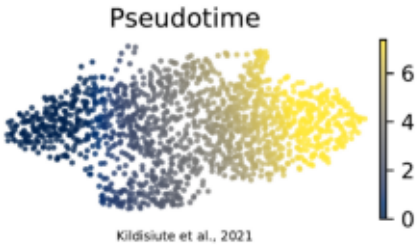
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

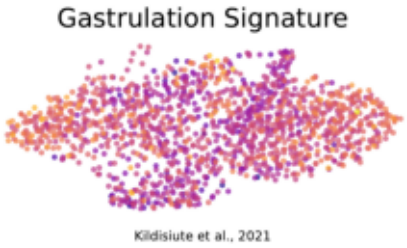
Gate 1: integration



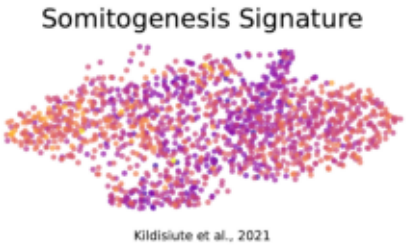
Trajectory / root



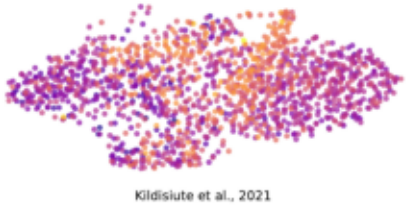
Gast



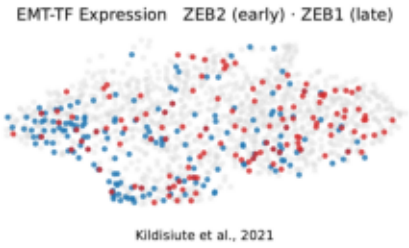
pSom



Crest



EMT-TFs



Dataset. Neuroblastoma (Kildisiute et al., 2021), 10x + CEL-seq2 — 21 patients / 19,723 cells (master scope). Multi-platform with platform-exclusive patients, so the 10x (UMIcounts) leaf is resolved; malignant only: 1,766 load, 1,720 across 2 samples (≥101) enter preprocessing; Embryonic compartment (1,504).

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. The neuroblastoma malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (non-branching), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. Unlike Lee, the axis is confined to the non-cycling lobe (Gast/pSom anti-correlate under proliferation), so the trajectory is scoped to it (Kürten lobe-identity precedent). EMT-TFs ZEB2 (early/root) → ZEB1 (late/terminus); FOXC/PRRX/SNAI near the detection floor; end-separation weak (Δpt=0.25).

Parameter choices

Min cells/sample	≥101 (keeps 2/5; gap 425→24)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	10x leaf + Embryonic lobe	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = Gast/pSom-high, Crest-low	**Compartments**	Embryonic 1,504 / Cycling 216